

Signorini's Problem

Max Stuthmann

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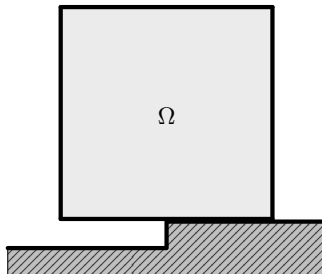
1 Introduction

2 Signorini's Problem

- Geometrical and Physical Setup
- Energy Minimization
- Weak Formulations
- Strong Formulations
 - A Hybrid Formulation
 - Classical Strong Formulation under Additional Displacement Regularity

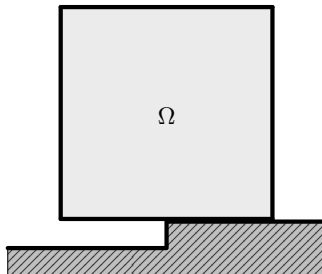
Rough Problem Setup

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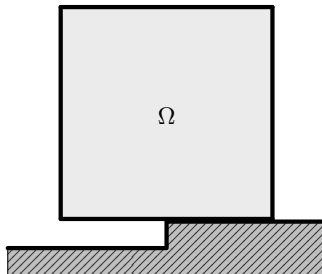
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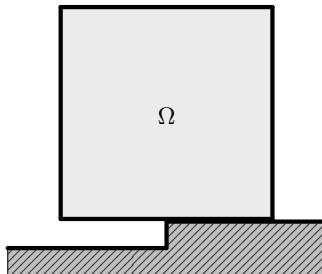
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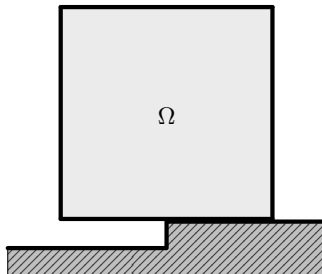
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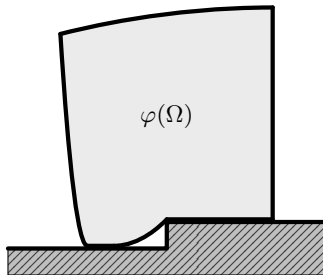
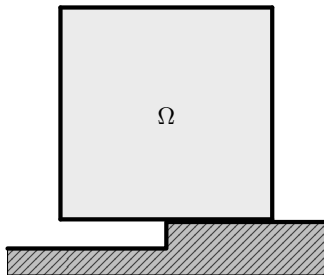
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- ▶ **Not in equilibrium configuration.**



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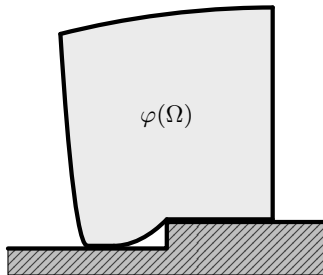
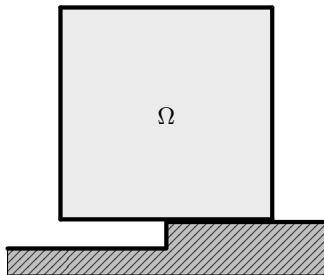
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$$\varphi(x) = x + u(x), \quad u : \bar{\Omega} \rightarrow \mathbb{R}^d.$$

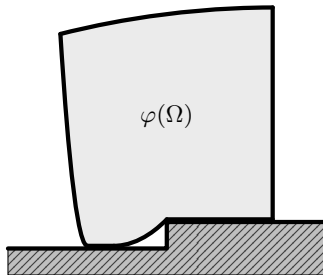
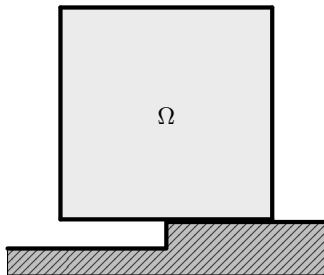


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- ▶ We aim to find the *displacement function* u .



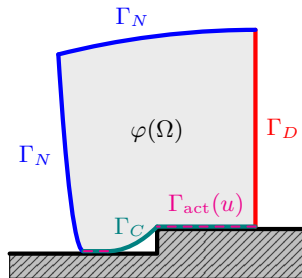
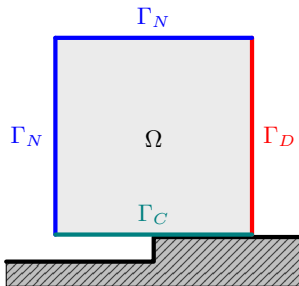
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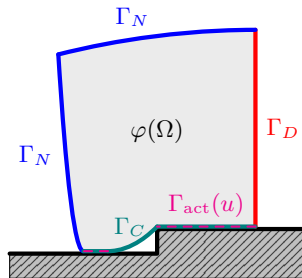
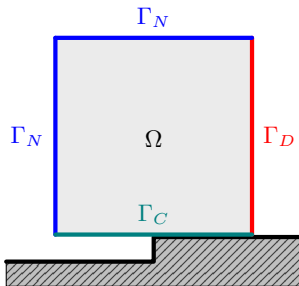
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- Decompose $\Gamma = \partial\Omega$ into three relatively open and pairwise disjoint parts $\Gamma = \overline{\Gamma_D} \dot{\cup} \overline{\Gamma_N} \dot{\cup} \overline{\Gamma_C}$ with $|\Gamma_D|, |\Gamma_C| > 0$ and $|\Gamma_N| \geq 0$.



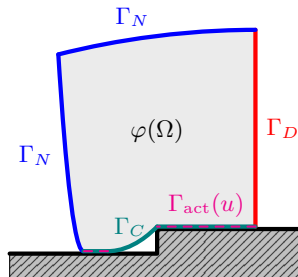
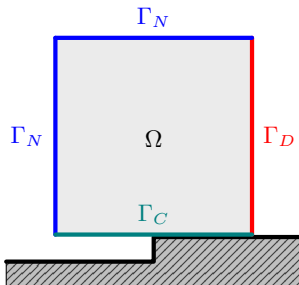
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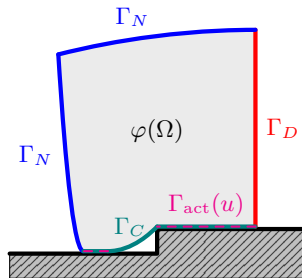
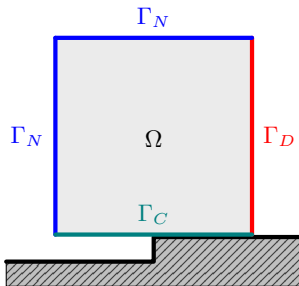
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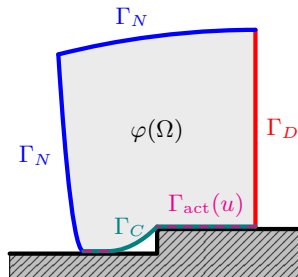
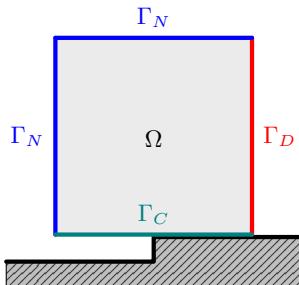
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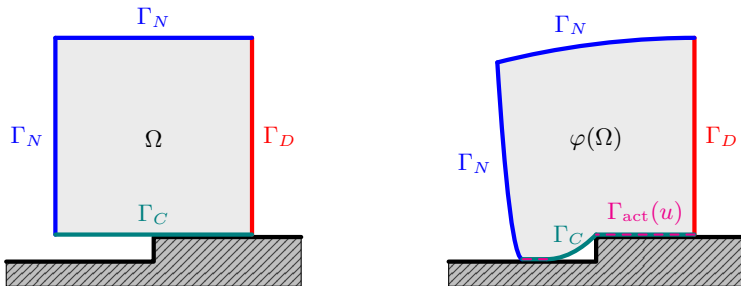
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- ▶ Rigid foundation can be describe by gap-function $g : \Gamma_C \rightarrow \mathbb{R}$. We assume $g = 0$.



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- If $\mathcal{A} : \Omega \rightarrow \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$ is a fourth order tensor and $s \in \mathbb{S}^d$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{kl}, \quad s : \mathcal{A}(x) : s := \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

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- ▶ *(Cauchy) stress tensor:* $\sigma : H^1(\Omega; \mathbb{R}^d) \rightarrow L^2(\Omega; \mathbb{S}^d)$, $u \mapsto \mathcal{A} : \varepsilon(u)$.

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- ▶ *Equilibrium* between both: $a(u, v) = \ell(v) \quad \forall v \in H^1(\Omega; \mathbb{R}^d)$.

Standard Properties of a and ℓ

Proposition

The symmetric bilinear form $a : V \times V \rightarrow \mathbb{R}$ defined above is

- i) *bounded, i.e. for all $v, w \in V$ there exists $C_1 < \infty$ such that*

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega}.$$

- ii) *V -elliptic, i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2.$$

Moreover, the linear functional ℓ defined above is bounded, i.e. for all $v \in V$ there exists $C_3 > 0$ such that

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega}.$$

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Basic Notations on Sobolev Spaces

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- ▶ $H^0(\Omega; \mathbb{R}^d) = L^2(\Omega; \mathbb{R}^d)$.
- ▶ *Trace Theorem*: Restriction of smooth functions $\overline{\Omega} \rightarrow \mathbb{R}^d$ onto the boundary $\Gamma = \partial\Omega \in \mathcal{C}^{m,1}$ extends uniquely to a continuous surjective linear trace operator

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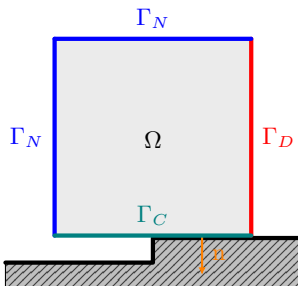
- ▶ For now, Ω is Lipschitz, i.e. $m = 0$ and we need $s = 1$, so

$$\gamma : H^1(\Omega; \mathbb{R}^d) \rightarrow H^{1/2}(\Gamma; \mathbb{R}^d)$$

Admissible Displacements and the Nonpenetration Constraint

- *Homogeneous Dirichlet* boundary conditions on Γ_D are incorporated in the underlying Hilbert space

$$V := \{v \in H^1(\Omega; \mathbb{R}^d) \mid (\gamma v)|_{\Gamma_D} = 0\}$$

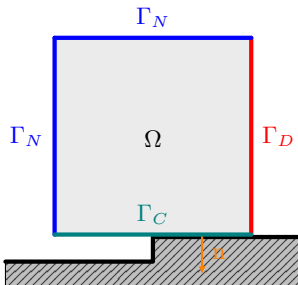


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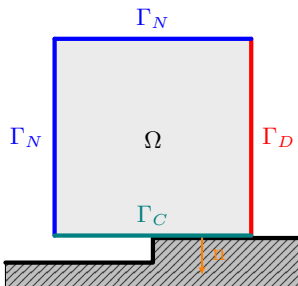


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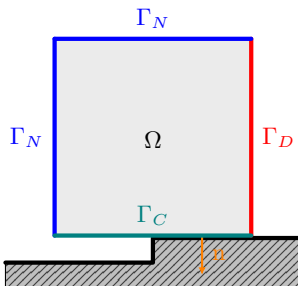


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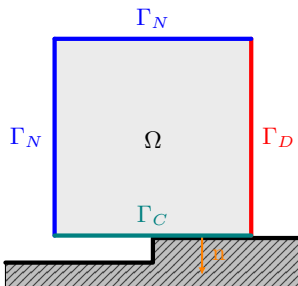


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- ▶ *Set of admissible displacements*: $K := \{v \in V \mid v_n \leq 0 \text{ a.e. on } \Gamma_C\}$.
- ▶ K is a closed and convex cone in V , i.e. if $v \in K$ then $\lambda v \in K$ for all $\lambda \in [0, \infty)$.



► *Equilibrium:*

$$a(u, v) = \int_{\Omega} \sigma(u) : \varepsilon(v) \, dx = \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} F \cdot (\gamma v) \, ds = \ell(v) \quad \forall v \in V$$

► *Equilibrium:*

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Problem (Signorini's Problem – Energy Minimization)

$$\text{Find } u \in K \text{ such that: } \mathcal{J}(u) = \min_{v \in K} \mathcal{J}(v).$$

Proposition (Well-Posedness of Signorini's Problem)

There exists exactly one $u \in K$ that minimizes $\mathcal{J}$ over K . Moreover, the unique minimizer satisfies the a-priori bound

$$\|u\|_{1,\Omega} \leq \frac{2C_3}{C_2} \|\ell\|_{V'}.$$

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- ▶ Stability bound follows from V -ellipticity of a w.r.t. C_2 and boundedness of ℓ w.r.t. C_3 .

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- Energy Minimization
- **Weak Formulations**
- Strong Formulations

Proposition (First-order optimality condition on convex sets)

A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality

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- ▶ Normal traces on $H(\operatorname{div}; \Omega)$ are understood as follows.

Normal Traces on $H(\operatorname{div}; \Omega)$

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- ▶ In the smooth case, $\gamma_{\mathbf{n}}(\tau)$ coincides with the classical traction $\tau \mathbf{n}$. We thus write $\tau \mathbf{n} = \gamma_{\mathbf{n}}(\tau)$.

Stress Regularity Assumption

- ▶ Thus, we further assume $\sigma(u) \in H(\operatorname{div}; \Omega)$ which yields $\sigma(u)_n := \gamma_n(\sigma(u)) \in H^{-1/2}(\Gamma; \mathbb{R}^d)$.

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- ▶ *Useful identity:* If $-\operatorname{div} \sigma(u) = f$ almost everywhere in Ω , then

$$a(u, v) - \ell(v) = \langle \sigma(u)n, \gamma v \rangle_\Gamma - \int_{\Gamma_N} F \cdot (\gamma v) \, ds, \quad \forall v \in V.$$

A Hybrid Formulation

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$, find $u \in K$ such that $\sigma(u) \in H(\text{div}; \Omega)$ and

$$\begin{cases} -\text{div } \sigma(u) = f & \text{a.e. in } \Omega, \end{cases} \quad (1a)$$

$$\begin{cases} \langle \sigma(u)n, \gamma u \rangle_\Gamma = \int_{\Gamma_N} F \cdot (\gamma u) \, ds, \end{cases} \quad (1b)$$

$$\begin{cases} \langle \sigma(u)n, \gamma v \rangle_\Gamma \geq \int_{\Gamma_N} F \cdot (\gamma v) \, ds \quad \forall v \in K. \end{cases} \quad (1c)$$

Equivalence of Hybrid and Weak Formulation

Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\operatorname{div}; \Omega)$. Then $u \in K$ solves the hybrid problem above if and only if it solves the weak problem

$$\text{Find } u \in K \text{ such that} \quad a(u, v - u) \geq \ell(v - u) \quad \forall v \in K.$$

Proof: *Hybrid $\Rightarrow$ Weak:*

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Proof: Hybrid $\Rightarrow$ Weak:

- ▶ Subtract (1b) from (1c):

$$\langle \sigma(u)\mathbf{n}, \gamma(v - u) \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma(v - u) \, ds \geq 0 \quad \forall v \in K.$$

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$$\text{Find } u \in K \text{ such that} \quad a(u, v - u) \geq \ell(v - u) \quad \forall v \in K.$$

Proof: Hybrid $\Rightarrow$ Weak:

- ▶ Subtract (1b) from (1c):

$$\langle \sigma(u)n, \gamma(v - u) \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma(v - u) \, ds \geq 0 \quad \forall v \in K.$$

- ▶ Insert the identity from the above:

$$a(u, v - u) - \ell(v - u) \geq 0 \quad \forall v \in K,$$

i.e. the weak formulation.

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► Let $\varphi \in \mathcal{D}(\Omega; \mathbb{R}^d)$. Then $u \pm t\varphi \in K$ for $|t| \ll 1$, hence the weak inequality implies

$$a(u, \varphi) = \ell(\varphi) = \int_{\Omega} f \cdot \varphi \, dx = \langle f, \varphi \rangle \quad \forall \varphi \in \mathcal{D}(\Omega; \mathbb{R}^d).$$

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- ▶ Thus, $-\operatorname{div} \sigma(u) = f$ in $\mathcal{D}'(\Omega; \mathbb{R}^d)$ and by $\sigma(u) \in H(\operatorname{div}; \Omega)$, it holds in $L^2(\Omega; \mathbb{R}^d)$ as well.

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- ▶ This is exactly (1b).

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1 Introduction

2 Signorini's Problem

- Geometrical and Physical Setup
- Energy Minimization
- Weak Formulations
- Strong Formulations
 - A Hybrid Formulation
 - Classical Strong Formulation under Additional Displacement Regularity

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- ▶ With $\sigma(u) = \mathcal{A} : \varepsilon(u)$, the stress tensor is now an operator $\sigma : H^s(\Omega; \mathbb{R}^d) \rightarrow H^{s-1}(\Omega; \mathbb{S}^d)$.

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- ▶ Hence the traction $t(u) := (\gamma\sigma(u)) n \in H^{s-3/2}(\Gamma; \mathbb{R}^d)$ is well-defined and if $\sigma(u) \in H(\operatorname{div}; \Omega)$ we have

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- ▶ Its normal and tangential components are given by

$$\sigma_n(u) := t(u) \cdot n \in H^{s-3/2}(\Gamma; \mathbb{R}), \quad \sigma_t(u) := t(u) - \sigma_n(u) n \in H^{s-3/2}(\Gamma; \mathbb{R}^d).$$

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- ▶ Furthermore, for $u \in H^s(\Omega; \mathbb{R}^d)$ we have $\gamma u \in (H^{s-1/2}(\Gamma; \mathbb{R}^d))$ with normal/tangential splitting

$$u_n := (\gamma u) \cdot n \in H^{s-1/2}(\Gamma; \mathbb{R}) \quad \text{and} \quad u_t := \gamma u - u_n n \in H^{s-1/2}(\Gamma; \mathbb{R}^d).$$

Classical Strong Formulation under Additional Displacement Regularity

Problem (Signorini's Problem – Classical Strong Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$ find $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ such that the small strain elasticity equations

$$\left\{ \begin{array}{ll} -\operatorname{div} \sigma(u) = f & \text{in } \Omega, \end{array} \right. \quad (2a)$$

$$\left\{ \begin{array}{ll} u = 0 & \text{on } \Gamma_D, \end{array} \right. \quad (2b)$$

$$\left\{ \begin{array}{ll} t(u) = F & \text{on } \Gamma_N, \end{array} \right. \quad (2c)$$

and the contact conditions

$$\left\{ \begin{array}{ll} u_n \leq 0 & \text{on } \Gamma_C, \end{array} \right. \quad (3a)$$

$$\left\{ \begin{array}{ll} \sigma_n(u) \leq 0 & \text{on } \Gamma_C, \end{array} \right. \quad (3b)$$

$$\left\{ \begin{array}{ll} \sigma_n(u) u_n = 0 & \text{on } \Gamma_C, \end{array} \right. \quad (3c)$$

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hold.

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Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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$$\begin{cases} -\text{div } \sigma(u) = f & \text{a.e. in } \Omega, \end{cases} \quad (4a)$$

$$\begin{cases} \langle \sigma(u)n, \gamma u \rangle_\Gamma = \int_{\Gamma_N} F \cdot (\gamma u) \, ds, \end{cases} \quad (4b)$$

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- ▶ Now, $t(u) := (\gamma \sigma(u))n \in H^{s-3/2}(\Gamma; \mathbb{R}^d)$ is smooth enough for both concepts to coincide:

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► With $t(u) = \sigma_n(u)n + \sigma_t(u)$ and $\gamma v = v_n n + v_t$, we have

$$t(u) \cdot (\gamma v) = \sigma_n(u) v_n + \underbrace{\sigma_t(u) \cdot v_t}_{=0 \text{ on } \Gamma_C \text{ since } \sigma_t(u)=0 \text{ on } \Gamma_C}.$$

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- ▶ It remains to prove $t(u) = F$ on Γ_N and the remaining contact conditions.

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$$\int_{\Gamma_N} t(u) \cdot g \, ds \geq \int_{\Gamma_N} F \cdot g \, ds, \quad - \int_{\Gamma_N} t(u) \cdot g \, ds \geq - \int_{\Gamma_N} F \cdot f \, ds.$$

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- ▶ Therefore,

$$\int_{\Gamma_N} (t(u) - F) \cdot g \, ds = 0 \quad \forall g \in H_{00}^{1/2}(\Gamma_N; \mathbb{R}^d) \quad \implies \quad t(u) = F \text{ on } \Gamma_N.$$